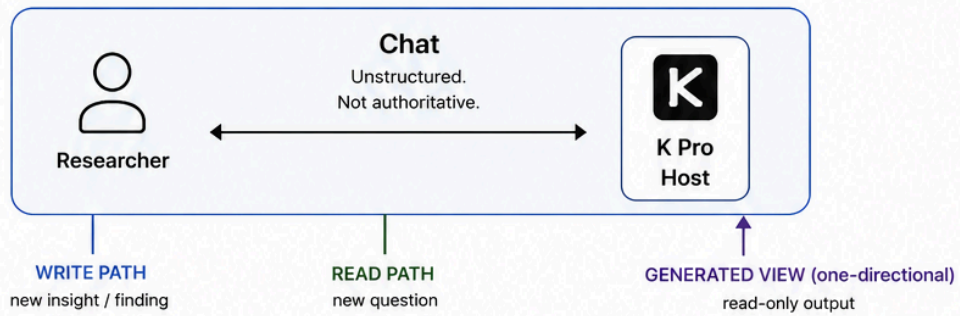
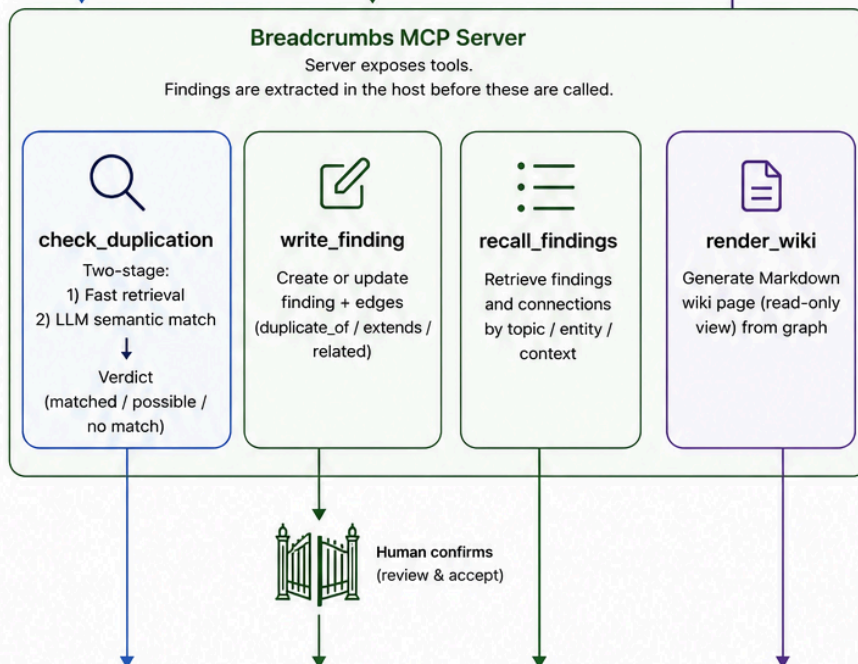


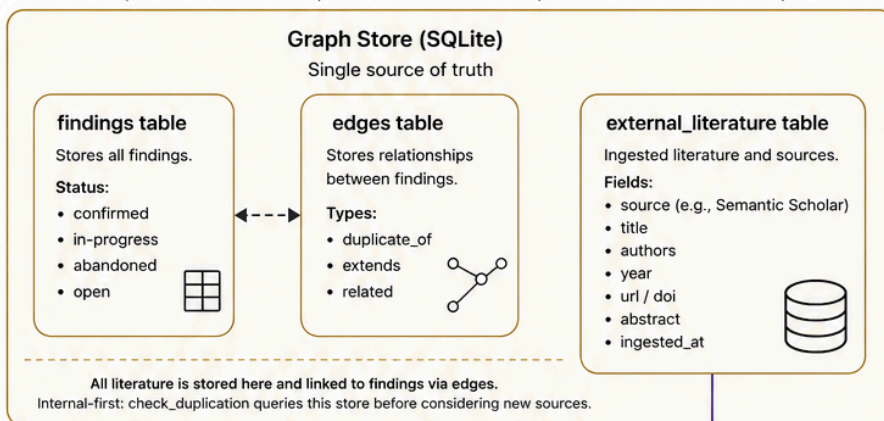
1 CHAT LAYER



2 BREADCRUMBS MCP SERVER



3 GRAPH STORE (SOURCE OF TRUTH)



- **Write Path** (create knowledge)
- **Read Path** (find what exists)
- **Generated View** (one-directional)
- 🚪 **Human Gate** (confirmation)
- **Tool / Component**
- **Persistent Store (SQLite)**

Markdown Wiki Page
Generated view.
Read-only.
Never edited back.

NOTES

- Extraction of findings happens in the host before tools are called.
- Internal-first: check_duplication queries the graph store (including ingested literature) before considering new sources.
- All authoritative data lives in the graph store.
- The wiki is a generated, read-only view.

Repo-

<https://github.com/MangosteenMadness/Breadcrumbs>

Pitch Deck:  breadcrumbs-pitch.pptx

The Challenge

Participants will develop MCP tools, agents, and workflows that extend the capabilities of AI scientist systems like K Pro.

How to submit

- **Team size:** min =2, max = 5
- **Submit your creations by 4pm Sunday July 12th using this project submission template**
- **Minimally**
 - Demo video <5min (mp4, mov only)
- **Preferred**
 - Demonstration of a working system / service
 - Open github repo
 - Pitch deck or 1 pager (Google Docs / Google Slides OR Microsoft Word / Powerpoint only)

Research is Hard

Ask question- Has this been asked before- Internally first and externally second

One sentence Pitch-

The one sentence

An internal research-memory layer for an AI-scientist platform (K Pro): before a researcher runs a hypothesis, it checks — **internally first** — whether someone in their own org already explored it (including *abandoned* attempts), then whether the published world has. External novelty is commoditized (FutureHouse's Owl does it). **Internal, in-flight memory is the wedge nobody owns.**

her asks a survival-stratification question on TCGA LUAD (lung adeno). Agent runs it, gets a real result (e.g., "high [signature] → worse OS, HR ~1.8, n=230"). Agent writes it to the graph with provenance (who, when, data, effect) and a **calibrated** novelty note ("no prior internal work; published record shows related work in X").

Session 2 — the save. A *different* researcher (new session) asks a related question — e.g., the same signature in LUSC (lung squamous). Instead of cold-starting, the agent responds:

"Before you run this — someone in your org tested a version of this in Session 1 [here's who, here's what they found]. Separately, a teammate **abandoned** a related approach because [reason]. Published record adds [context]. The genuinely open part is [X]."

The punchline: internal recall + an abandoned-attempt surfaced = something Owl and the consulting knowledge graphs structurally cannot do, because both only see the *published* record where failures never appear.

Requirements

P0 — must work or there's no demo

1. **Graph store** with findings as nodes: {*disease, hypothesis_text, signature, effect, n, status, author, timestamp, provenance*}. Status ∈ {confirmed, in-progress, **abandoned**}.
2. **write_finding** — append-only, provenance-tagged.
3. **recall_findings** — given a new question, return semantically-related prior findings (LLM-matched).
4. **One real analysis** — TCGA slice → survival stratification → produces a finding object. Thin and real.
5. **The two-session flow** runs end-to-end at least once.

P1 — makes it win (add once P0 is solid)

6. **check_duplication** presenting **internal first, external second**.
7. **External literature check** — PubMed/Europe PMC API call (free), with cached fallback.
8. **Abandoned-result surfacing** as a first-class result type. *This is the differentiator — pre-seed 1–2 abandoned findings with reasons.*
9. **Calibrated language** everywhere: "no prior work found in [sources]," never "this is novel."

P2 — only if hours remain

10. Cross-indication recall shown as a graph visual.
11. A "novelty verdict" badge attached to each stored finding.

Technical architecture (lightest path)

- **MCP server (Python + FastAPI).** The deliverable *is* an MCP server — this reinforces the "add-on to K Pro, consumed via MCP" thesis. Tools: `run_analysis`, `write_finding`, `recall_findings`, `check_duplication`.
- **Graph store:** SQLite + a thin relational schema, or `networkx` in-memory persisted to JSON. **Do not stand up Neo4j** — setup cost isn't worth it in 13h.
- **Semantic matching:** Claude API — "are these two hypotheses the same question?" No embedding infra needed.
- **External check:** Europe PMC REST (no key) or PubMed E-utilities. Cache results so the demo can't be broken by a flaky network.
- **Data:** pre-stage one indication's **clinical + bulk RNA** before 12:00 if possible. cBioPortal API is the fastest way to pull TCGA slices. You need enough to run one real Kaplan-Meier/Cox, not all 10K patients.
- **Demo surface:** cleanest option — connect the MCP server to **Claude Desktop** and drive it live (the MCP host stands in for K Pro). **Record a backup video** of the full flow tonight — do not trust demo-gods at 16:00. Fallback: a 1-page thin chat UI (Next.js) you fully control.

Roles

Person	Owns
Biologist 1	Locks the disease + the ONE hypothesis. Picks the signature. Writes the biological demo script.
Biologist 2	Seeds the graph with realistic prior findings — incl. 1–2 abandoned ones with reasons. Live-validates that recalled matches are credible. On-stage "this is a real match" voice.
Builder 1	Data + analysis pipeline: TCGA slice → survival stratification → finding object. <code>run_analysis</code> tool.
Builder 2 (stronger)	The moat: graph store + <code>write_finding/recall_findings/check_duplication</code> + LLM semantic match + external lit check.

Product/Pitch (Yousuf) Agent orchestration glue, demo surface, scope enforcement, deck, narrative, Owkin positioning.

The demo (write everything toward this)

Session 1 — a finding is born. Researcher asks a survival-stratification question on TCGA LUAD (lung adeno). Agent runs it, gets a real result (e.g., "high [signature] → worse OS, HR ~1.8, n=230"). Agent writes it to the graph with provenance (who, when, data, effect) and a **calibrated** novelty note ("no prior internal work; published record shows related work in X").

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"Benchling Lab Data -

The punchline: internal recall + an abandoned-attempt surfaced = something Owl and the consulting knowledge graphs structurally cannot do, because both only see the *published* record where failures never appear.

We use LLM wiki style. Log and content so